

Square Subsets

Time limit: 1.00 s

Memory limit: 512 MB

Given an array of n integers, count the number of subsets whose elements' product is a square number. Also count the empty subset with product equal to one.

Input

The first line has an integer n : the size of the array.

The next line has n integers $x_1, x_2, \dots, x_n$: the contents of the array.

Output

Print one integer: the answer to the problem modulo $10^9 + 7$.

Constraints

- $1 \leq n \leq 5000$
- $1 \leq x_i \leq 5000$

Example

Input	Output
4 2 2 3 6	4